

Its inspiration, responsive architecture, uses robotics and motion sensors to accomplish the task. Web design requires a very abstract way of thinking and involves the use of different technologies. Certain technologies and ideas that are already being used by Web developers are fluid layouts, media queries and scripts, CSS, etc, in order to reformat Web pages automatically.

Do not for a moment mistake this approach as being only about adjustable screen resolutions and automatically resizable images. It is, in fact, a completely revolutionary way of thinking about Web design.

Let's look at some of its features and a few that are still in the process of evolution.

Adjustable screen resolutions

With every new day bringing up a new device with a new screen resolution, definitions and orientations, the question is: how does one design a website that fits in the portrait and landscape format at the same time. Flexible layouts seem to solve this problem to an extent.

As the name suggests, these layouts keep everything flexible according to viewing requirement. While this approach is not a complete fix, it's a perfect solution for the devices that support shifting from portrait to landscape orientation, instantly.

The Web page in Figure 1, gives readers a perfect idea of how flexible layouts can be useful. The design in this figure can be created using fluid grids, fluid images and smart markups, wherever needed. To know more on how to create fluid grids and fluid images, check out the Reference section at the end of the article.

Flexible/fluid images

One of the major issues in responsive Web design is working with images. The idea behind fluid images is to deliver images in the maximum size they will be used in. Here, we don't

declare the size of the image in the code but let the browser resize the image as needed, while using CSS. There are many techniques for resizing images accordingly. Let's look at some of the most popular options.

CSS's 'max-width' (`img {max-width: 100%;}`): It works as long as no other style overrides it. It makes sure that the image loads to its maximum size unless the viewing area is smaller than the image size. Here, '100 per cent' means 100 per cent of the screen and browser width and when that 100 per cent is smaller, the image shrinks as well. However, 'max-width' is not supported in Internet Explorer.

Responsive images: The issue with the above approach is the image download time. Because, if the original image is designed for larger resolutions and even if it is resized for a mobile device, the download size still remains the same and may be slow.

Responsive images solve this problem by shrinking the image resolution on smaller devices. This implementation requires a little extra effort to create a few more files.

A JavaScript (`rwd-images.js`), the `.htaccess` file and the image file can all be put in an HTML tag to reference both smaller and larger resolution images:

```

```

This will download and load the 'largeRes' image for any screen larger than 480 pixels; otherwise, it will load the 'smallRes' image.

To see the content or download all the files needed, refer to the GitHub location in the Reference section.

Stopping iPhone from resizing the images: One feature of iPhone is to resize the images automatically to fit to the screen. Hence, with the introduction of flexible images, iPhone still adjusts the images even when they are delivered in optimal size. A workaround is to use an Apple-specific meta tag. Place this tag below the `<head>` section and you are done.

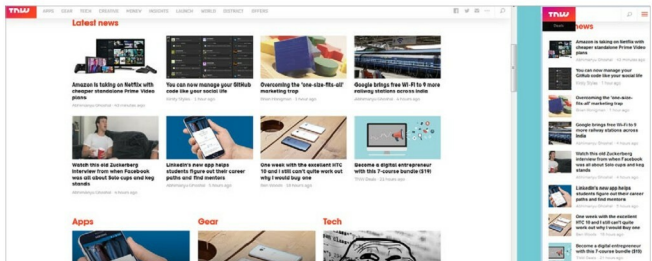


Figure 1: Flexible layout: landscape to portrait